Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Build a Human-AI Workforce with Autonomous AI Agents

TGS Ref No: TGS-2024043854

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 3.9

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	12 July 2026	Initial release — 3-day lesson plan for the autonomous-AI-agent labs (Hermes Agent, OpenClaw, Paperclip).	Dr. Alfred Ang
1.1	14 July 2026	Revised to a 2-day plan (Day 1: Topics 1 & 2; Day 2: Topic 3 & final assessment). Hermes expanded to 14 labs, OpenClaw to 12, Paperclip to 8 (34 labs total). Learning outcomes aligned to the Course Proposal / TSC; per-lab reference videos added.	Dr. Alfred Ang
1.2	14 July 2026	Visual redesign: all content slides card-format; YouTube links removed from the deck and Learner Guide (videos remain in the labs); live Hermes screenshots and skill-category grids added; Hermes install lab expanded with the official install methods.	Dr. Alfred Ang
1.3	14 July 2026	Deck restyled in the dark masterclass theme end-to-end; added the Hermes masterclass slides (What is Hermes; The Bet: Compounding Value; platform matrix; Top-10 slash commands; Telegram setup; memory L1-vs-L2 and session-search; what-is-a-tool, built-in tool catalog, tools-in-action, cron scheduling, delegation, kanban orchestration and OpenClaw topic slides); Lab 13 rebuilt as the multi-agent video team (profiles + kanban) with detailed Labs 11-14 guides; fade slide transitions and the live Achievements dashboard screenshot.	Dr. Alfred Ang
1.4	15 July 2026	Security masterclass slides added to the Hermes Security lab (defense-in-depth layers; the gateway's six-check default-deny trust chain; channel lockdown dials; the three approval modes + the Tirith scanner; a live Dangerous-Command approval; YOLO mode; the unrecoverable-command blocklist floor; prompt-injection scanning of context files). Security lab steps expanded with the channel allowlist and approvals.mode.	Dr. Alfred Ang
1.5	15 July 2026	Sixteen concept diagrams imported from the original Mastering OpenClaw masterclass deck into Topic 2 (OpenClaw OS control plane; deployment; hub-and-spoke gateway; tools vs skills; SKILL-on-demand; 6-phase execution loop; exec; HEARTBEAT + autonomic tick; workspace file persistence; context assembly; compaction; defense methods; poisoned-skill anatomy; sessions_spawn; sub-agent routing).	Dr. Alfred Ang
1.6	15 July 2026	Topic 3 (Paperclip) rebuilt as the 'Tertiary AI News Research' company across 6 labs (27-32): install on Windows & Mac; company, CEO & mission; model adaptors; a detailed task backlog whose core task hires the team; the Tavily Search API via Secrets; and hiring the members under the hiring task — with live dashboard screenshots. Natural-language prompt steps added to the Hermes and OpenClaw labs alongside the CLI. Platform logos added to the Three Platforms page. 32 labs total.	Dr. Alfred Ang
1.7	15 July 2026	Sixty-eight content slides migrated from the original 'Mastering OpenClaw' masterclass deck into Topic 2 as framed figures in their matching labs — agent evolution and Agent = Model + Harness, the OpenClaw Wikipedia history and OpenAI acquisition, deployment options and services, model stacks and Mission Control, channels, skills and ClawHub risks, tools (AgentMail, Agent Browser, Firecrawl), system prompt / workspace / context and memory, security lockdown, slash commands, crons, and use cases (NFT certificates, dashboard, cost saving, Google integration).	Dr. Alfred Ang
1.8	15 July 2026	Imported masterclass figures re-blended into the dark deck	Dr. Alfred Ang

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		theme: original title and copyright strips cropped, white canvases converted to the deck background with text inverted to ivory (embedded screenshots untouched), frameless rendering, and duplicated titles reworded.	
1.9	16 July 2026	Deck curated per trainer review: the bulk masterclass figure imports were removed, keeping nine key figures (Wikipedia history, Agent=Model+Harness, SKILL.md, ClawHub, AgentMail, 10-steps lockdown, HEARTBEAT, autonomic tick, workspace persistence) re-exported at high resolution, with the Topic 2 opening story reordered ahead of Key Concepts. Every CLI step now also shows the natural-language prompt alternative ('Or just ask') across the Hermes and OpenClaw labs, in the deck and Learner Guide.	Dr. Alfred Ang
1.10	16 July 2026	PP assessment updated to v4 — evidence screenshots changed to: Task 1 Hermes interacting on Telegram or the multi-agent Kanban board; Task 2 OpenClaw on Telegram or the TUI; Task 3 the Paperclip setup (company, mission, CEO, hired agents) — with Topic 3 lab references refreshed. A 'RAG vs a Standard LLM Assistant' concept slide added to Core Concepts to ground WA Q2. Final PPT/LP/LG/labs/assessment alignment review.	Dr. Alfred Ang
1.11	16 July 2026	Topic 3 company renamed to 'Altera AI Blogs' with the mission re-pointed at publishing AI-related blogs end to end; the core backlog task is now 'Hire the agents to publish the AI-related blogs end to end' and Lab 32's hiring happens explicitly under it; Lab 29 (Adaptor) and Lab 32 descriptions expanded in full; NEW Lab 33 'Routine & Trigger — Publish Daily at 3pm' (33 labs total), verified with a live 'Run now' pipeline test and two live screenshots.	Dr. Alfred Ang
2.0	18 October 2026	Major revision. Deck redesigned in the 'Agentic AI Applications with Claude Code' house design. Topic 1 rebuilt as Foundations: history of AI, AI agents vs agentic AI, the agentic loop, harness engineering, the brain–perception–action framework, short/long-term memory, world models, single → multi-agent → human–agent collaboration paradigms, agent society, the Oct 2026 agent market (OpenClaw, Hermes Agent, Paperclip, OpenAI Dots, Meta Muse, Block Buzz, Manus Cue and more), Singapore's AgentSea (12,000+ agents) and human–AI use cases in science, maths, drug discovery and business. OpenClaw labs removed; 33 feature labs replaced by 10 connected use-case labs for the fictional Kopi Kaki Co. — Day 1 Hermes Agent (research brief, end-to-end social media campaign, human-in-the-loop publishing, multi-agent video studio, campaign command centre) and Day 2 Paperclip (found the company, hire the team, launch project, weekly video routine, govern & improve) — with mock data, brand assets, skills, instruction PDFs and a local Lab Prompts page. Assessment v5 aligned; FG and AP added.	Dr. Alfred Ang
2.1	29 October 2026	Human–AI landscape rebuilt with practical workflows, research evidence, collaboration modes and humanoid futures; Day 1 single Hermes office automation with gog, Excel, interactive dashboard and professional PPT; Day 2 Kanban and Claude Code video team; SWDA subscription steps and connected GitHub Pages prompts.	Dr. Alfred Ang
2.2	29 October 2026	Streamlined to eight connected activities, four per day, using Hermes Desktop and OpenAI Dots. Removed Paperclip and Claude-specific training. PP v6 evidence: Telegram, single-agent professional PPT with skills/tools, and multi-agent Kanban video creation with Hermes bots. FG/AP v5 aligned.	Dr. Alfred Ang

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		Corrected shared use-case slide spacing.	
2.3	29 October 2026	Day 1 reordered: brief AI history, two human–AI work modes, always-on agents, harness and Big Four comparison using BULDRR plus official product documentation. Added Meta Muse Singapore launch news (27 October), official app links and one accessible assistant setup activity. Company scenario moved to afternoon Hermes Desktop/skills labs; Google/Excel, dashboard/PPT, single-agent video and approved test email. Day 2 kept. PP v7: Task 1 two images (chosen assistant and Hermes artifact), Task 2 Activity 4 PPT; FG/AP v6 aligned.	Dr. Alfred Ang
2.4	29 October 2026	Admin slides aligned with the 'Agentic AI Applications with Claude Code' deck: 'Login to LMS and Download Course Material' slide with the current lms-tms.tertiaryinfotech.com sign-in screenshot replaces the outdated LMS slide; Claude Certified Associate (Foundations) badge added to Dr. Alfred Ang's trainer profile. Always-on agents: the four generic 'product inspection' slides replaced by step-by-step 'How to Set Up' walkthroughs for Claude Duet, Meta Muse, Grok Bot and OpenAI Dots from each vendor's official documentation (duplicate route slides merged). AI history timeline: Language era 2017–2023, Action era 2024 onward. Product-development slide rewritten as a worked Kopi Kaki Co. launch example (agent work vs human decision per stage). Whole deck re-sequenced for one continuous narration: Topic 1 runs history → agentic AI → human–AI work and use cases → always-on and the Big Four → Lab 1 setup walkthroughs and checks → Lab 1 → 'Under the hood' (the harness slides merged into one post-lab sequence, 40 min); Topic 2 introduces Kopi Kaki and its data, then Hermes, then places each lab's concept slides directly before that lab (Desktop and skills → Lab 2; Google Workspace and Excel → Lab 3; dashboard, PPT, video and test email → Lab 4) and closes Day 1 with reflection; Day 2 concept slides now precede Labs 6–8 as for Lab 5. Duplicate 'One Agent, Four Office Skills' slide removed. LP sessions and slide ranges re-mapped; LG Topic 1 re-ordered to match, with the product-development example and the four setup walkthroughs added. Timeline circles size to fit their words. QA fixes: Lab 4 now shows its video and approved-email step slides; lab overview cards and topic-divider subtitles no longer overflow; Lab 4 has its own subtitle; Hermes architecture slide links back to the morning harness; LP lab-reference and FG briefing slide ranges corrected; WA Q6 answer cites Activities 5 and 8.	Dr. Alfred Ang
2.5	29 October 2026	Always-on comparison extended to the Big Five with Manus Cue (official Manus sources). A features slide for each agent (Claude Duet, Meta Muse, Grok Bot, OpenAI Dots, Manus Cue), plus Manus general-agent capabilities with worked examples and the Manus 2.0 additions; Big Five comparison follows the features. 'How to Set Up Manus Cue' added as a reference route (assessed routes unchanged). Practice Exam slide (exams.tertiaryinfotech.com) restored before the final assessment. LG Topic 1 and LP aligned.	Dr. Alfred Ang
2.6	29 October 2026	Streamlined for one continuous narration (223 → 158 slides). Topic 1 now runs history → what agentic AI is → under the hood (moved before Activity 1) → human-in/on-the-loop → work modes and use cases → always-on agents → Activity 1. One briefing slide per always-on agent (Muse, Dots, Grok Bot, Duet, Cue: features plus setup) replaces the separate feature, setup and Singapore-news slides. Every topic now ends with	Dr. Alfred Ang

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		its lab, and Day 2 opens on the Topic 3 divider, which carries the Day 1 recap; overlapping slides were merged or removed (e.g. duplicate skill/tool/permission checks, Day 1 close, Day 2 outcome, repeated RAG and Dots-control slides). Day 1 morning re-timed within LU1; LU hours, activities and assessment unchanged.	
2.7	30 October 2026	Trainer review of Topic 1 (158 → 145 slides). Added a Singapore news slide after the AI history (Synapxe AgentSea: 12,000+ agents created by 80,000+ public healthcare professionals; The Straits Times, 24 Aug 2026). Added an AI-for-Science news slide after AI for Science (Navier–Stokes, Stanford virtual biotech, 700 OpenAI preprints, OpenAI maths/physics/computing claims), each with its verification status. Added five collaboration-pattern slides (one human/one agent, multiple humans/one agent, one human/multiple agents, multiple humans/multiple agents, inter-company), each with an illustration, an explanation and, for patterns 1–3, a real screenshot with third-party names blurred. AI for Science and the humanoid elderly-care task now show their illustrations beside numbered steps. Each Big Five briefing slide carries the product logo; YouTube Shorts for Muse, Dots, Grok Bot, Duet and Cue embedded as PowerPoint online videos that play on the slide (poster frame and link in the PDF). Removed in Topic 1: application-categories table, the detailed harness slides (weights vs context, context assembly, memory vs knowledge base, RAG, standard LLM vs RAG, authority boundary, four checks, harness limits), human-in/on-the-loop, product development, always-on runtime, Manus general agent, the four Activity 1 check slides, the PP Task 1 slide and the Lab 1 checkpoint slide. The LG keeps the assessed content (application categories, context/memory/RAG, product development); WA answer-key citations re-bound to the remaining slides and LG; questions unchanged.	Dr. Alfred Ang
2.8	30 October 2026	Published release of the Topic 1 review. The use-case gallery is now 'AI Agents for Business: Five Use Cases' (customer service, IT service management, finance, sales, procurement and supply chain; Genic Solutions, 30 Sep 2026), replacing the 'Two modes' gallery. AI-for-Science news slide (Navier–Stokes, Stanford virtual biotech, 700 OpenAI maths preprints, OpenAI maths/physics/computing claims) with verification status. New visual analogy slide 'An AI Agent Works Like a Person' (brain + body → food → work; model + harness → tokens → work); the under-the-hood hub now has 'AI agent' at its centre with the model as one component. QA fixes: LP column widths and Lab 1 checkpoint slide range, LG heading/image pagination, Muse and Cue briefing polish, answer-key wording.	Dr. Alfred Ang
2.9	30 October 2026	The five collaboration patterns are now five collaboration scenarios. Each slide has a one-line pitch, a vivid hook and three named real-world examples (e.g. Scenario 1: OpenAI Dots, Meta Muse, Grok Bot, Claude Duet, Manus Cue; Hermes Desktop; Claude Code/Codex). The repeated 'How it works / The human decides / Watch for' rows were removed. LG 1.5 rewritten to match.	Dr. Alfred Ang
3.0	30 October 2026	Trainer review of v2.9. Activity 1 is now two parts: 1A, every learner joins Meta Muse at muse.ai (Facebook sign-in) and redeems the trainer's invite code H1KWDZ in Settings → General → Usage within 48 hours (1 billion Muse tokens each); 1B, learners whose ChatGPT plan includes dots (Pro and Business Premium at launch, per OpenAI) set up an OpenAI	Dr. Alfred Ang

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		Dot with a standing job. Two new visual lab slides (invite card, plan check, embedded Dots setup video). Grok Bot moved to a reference route. New slide after Scenario 2: learners scan Tertiary Infotech Academy's WhatsApp Business QR code and try a one-agent, many-people chat live. Assessment unchanged.	
3.1	30 October 2026	YouTube videos now play inside PowerPoint: each video carries PowerPoint's online-video embed (web video iframe) instead of a linked media file. Activity 1 now uses the Kopi Kaki Co. scenario: learners check any company meeting (launch review, supplier tasting, outlet visit) and draft a reply; the 'Meet Kopi Kaki Co.' slide moves from Topic 2 into Activity 1. Hermes Desktop download link added to its setup slide and the Learner Guide; the native-vs-WSL runtime slide removed. PP Task 1 wording: 'a meeting schedule' instead of 'a training schedule'; questions, K/A codes and rubric unchanged.	Dr. Alfred Ang
3.2	30 October 2026	YouTube videos fixed to play on click in PowerPoint: each video now uses the exact markup and click-to-play timing of PowerPoint's Insert › Video › Online Film, copied from a trainer-tested sample (v3.1 videos showed only the poster).	Dr. Alfred Ang
3.3	30 October 2026	Activity 1C added: learners install Wispr Flow (wisprflow.ai) and speak their prompts to Muse, Dots and Hermes instead of typing; new pitch slide with a typed vs spoken before/after and a download button. Activity 1B trimmed to 10 minutes; Lab 1 stays 75 minutes. Every video also gets an Open in YouTube fallback link. Trainer note: PowerPoint blocks online videos in a downloaded deck until Enable Content is clicked.	Dr. Alfred Ang
3.4	31 October 2026	One founder story: you are the founder-CEO of Kopi Kaki Co., a Singapore kopi startup, and AI agents help you start it — three mandatory topics and one optional bonus. Topic 1 unchanged (Activity 1: Meta Muse or OpenAI Dots as your always-on CEO assistant). Topic 2 rebuilt as 'Hermes Agent — Your First AI Employee': what Hermes is, Hermes Desktop, connecting a model (MiniMax demo; the one-day trial is marked awaiting verification), Telegram, tools, reviewed skills, Firecrawl web search, gog for Gmail/Calendar with one approved send, cron schedules and a daily 8:00 email-summary cron, with a guided setup lab; Activity 2 installs UX skills from skills.sh and builds the Kopi Kaki website; Activity 3 uses a PPT tool and skill to build the Kopi Kaki pitch deck (PP Task 2). Topic 3 rebuilt as 'Hermes Teams': a general introduction to multiple agents (each agent has its own tools, unique skills, model, instructions and context window), the trainer's three multi-agent slides recreated (Agent Communication Protocol, 'Each agent can…', the damaged-order hand-off), video skills and tools (HyperFrames, Remotion, Manim), profiles, Kanban and bots; Activity 4 social media marketing team (researcher, LinkedIn/Facebook/Instagram writers, image creator, posts gallery); Activity 5 video production team (researcher, scriptwriter, voice-over, producer, reviewer, human approval via a review link; PP Task 3); grounding/RAG and evaluation concepts; the 5 Principles of AI Security (as in the Agentic AI Applications with Claude Code deck). Bonus topic: Paperclip — Kopi Kaki as a one-person company (CEO with instructions → Chief of Staff → marketing team → org chart → the launch video, compared with the Hermes Kanban team), using the original Paperclip screenshots. Old Activities 3, 6, 7 and 8 removed. Mock company reframed as a startup. PP evidence:	Dr. Alfred Ang

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		Task 1 one Muse/Dots screenshot; Task 2 one Hermes PPT screenshot; Task 3 two screenshots (Kanban board + video); questions unchanged. Labs, lab PDFs, COPY-PROMPTS and the prompts page regenerated with every step and copy-ready prompt, including Paperclip.	
3.5	31 October 2026	No command lines for learners: every Topic 2 and Topic 3 step is now a click in Hermes Desktop or a prompt Mia carries out. Trainer screenshots of Hermes Desktop v0.22.5 added with explanations (version check, Settings → Model, Telegram Create with QR and Connected with the user ID blurred, Settings → Voice, Capabilities → Skills and Connectors, install a skill by asking, Scheduled jobs, bot Edit Profile, bot group chat, the Activity 2 website). Model: this class uses the learner's ChatGPT Plus plan (the free AI subscription slide now says to choose ChatGPT Plus); the Nous Portal Free tier ($0, free models only) and paid Nous plans (Plus USD 20, Super 100, Ultra 200) shown as alternatives; the MiniMax demo removed. Gmail now connects with a Google sign-in through Capabilities → Connectors (no OAuth client, no gog CLI; free-tier connector access marked awaiting verification); a reviewed kopi-email-brief skill replaces gog-office. Activities: 2 and 3 are explicitly single-agent; Activity 4 now hires five named bots (Rafi, Lin, Farah, Ivy, Pixel) with New Agent and runs them in one group chat; Activity 5 builds the video team and its Kanban board from one prompt (voice-over uses Hermes's free Edge text-to-speech); new Activity 6 schedules jobs that auto-generate the weekly social drafts and a video draft, still behind human approval. Day 2 retimed inside the approved LU3 4 hours (Lab 5B 70 min, Lab 6 30 min, Paperclip demo 20 min). Paperclip bonus installs through Mia. Assessment questions, K/A codes and PP evidence unchanged.	Dr. Alfred Ang
3.6	31 October 2026	Topic 1: the Big Five comparison is back with the model inside each agent (Muse Spark, GPT-6 Astra, Grok 4.6/4.7, Claude Opus 5.5, Manus 2.0 harness), what to use it for and a Kopi Kaki job, checked against each maker's announcement on 31 Oct 2026. Topic 3: the ACP/A2A protocol content is removed; the trainer's multi-agent screenshots are redrawn as native diagrams with Kopi Kaki examples: hub-and-spoke, how a request travels, task decomposition (the 5-step framework, answered for Activity 5) and subagents (Hermes delegate_task). Trainer hand edits ported: no Checkpoint & Reflection slides in the labs, no 'Was the team worth it?' slide. Topic 2 reordered by the trainer: what an AI agent is (components moved from Topic 1) → what Hermes Agent is → install, then hand Mia the AGENTS.md rulebook (+ → Folder… attaches the kopi-kaki folder; the first prompt tells her to follow it) → provider and model (Nous free Upstage Solar Mini 4 or ChatGPT Plus) → Telegram, two ways (Create with QR, or manually with BotFather) → tools and skills (the difference, moved from Topic 1; Gmail, web design and PPTX; Firecrawl kept for Topic 3) → Activities 2 and 3. Every lab slide links to its section of the Lab Prompts page. Assessment v7 unchanged.	Dr. Alfred Ang
3.7	1 November 2026	A prominent orange COPY THE PROMPTS button on every lab intro and prompt slide opens that lab on the Lab Prompts page (prompts/#lab-<code>). The Practice Exam slide is removed at the trainer's request. Learner Guide: the Topic 2 setup lab now sits before Activities 2 and 3, as in the deck. Assessment v7 unchanged.	Dr. Alfred Ang

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3.8	1 November 2026	Topic 1 Big Five comparison: the At Kopi Kaki row is replaced by Single or multi-agent (Muse and Dots: single agent only; Grok Bot and Manus Cue: multi-agent; Claude Duet: unknown), and a Pricing row is added (Muse: free; Dots, Grok Bot and Duet: paid; Cue: free during early access, checked 1 Nov 2026), in the deck and Learner Guide 1.7. Assessment v7 unchanged.	Dr. Alfred Ang
3.9	1 November 2026	Topic 3 Activity 5 shows the trainer's own run in Hermes Desktop: the five video agents in the sidebar, two Kanban screenshots of cards handing off (voice-over running, then render ready) and two frames of the rendered vertical how-to video, each explained on its slide. Learner Guide 3.4 and the Activity 5 lab section use the same screenshots. Topic 1 scenarios 1–3 each lead with a named use case: personal assistant, chatbot and customer service (with Klarna's AI assistant), and one-person company (with Paperclip); Learner Guide 1.5 matches. Assessment v7 unchanged.	Dr. Alfred Ang

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Course Information

Course Title	Build a Human-AI Workforce with Autonomous AI Agents
WSQ Course Reference	TGS-2024043854
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	9:30 am–6:30 pm both days. 16 scheduled course hours excluding lunch include 13 hours instruction/practice, 1 hour tea breaks and 2 hours assessment. Tea is included in the scheduled LU allocations; this is not a claim of 14 net instructional hours.
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led; always-on assistant setup followed by Hermes Desktop office and team practice (learner laptop)
Course scenario	Kopi Kaki Co. — a fictional Singapore kopi startup you found, launching the Gula Melaka Cold Brew at its Tiong Bahru flagship on 1 Dec 2026 (#ShiokColdBrew) — all company data is fictional
Slide deck	Human-AI Workforce with Autonomous AI Agents-v3.9.pptx (160 slides)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Analyse AI (LLM-based) applications across a range of industries to identify their capabilities and limitations.
- LO2: Establish the relationship between AI algorithm/agent design and chatbot/agent efficiency.
- LO3: Evaluate and improve the effectiveness of AI (RAG and agent) applications in product development.

Topics and Labs

Topic	Content / labs	Slides
Topic 1: Human–AI Collaboration & Always-On AI Agents	Brief AI history and what agentic AI is; Under the hood: the model in a harness, tools, skills and the agentic loop; humans decide; Two work modes, five collaboration scenarios, use cases and the Big Five always-on agents; Activity 1 · Join Meta Muse (invite code) and set up an OpenAI Dot. Activity 1: always-on assistant setup.	20–56
Topic 2: Hermes Agent — Your First AI Employee: Setup, Tools, Skills & Schedules	What an AI agent is, part by part; what Hermes Agent is; install Hermes Desktop and point it at your kopi-kaki folder; Provider and model: Nous Portal free (Upstage Solar Mini 4) or your ChatGPT Plus plan; Telegram bot by QR code or BotFather; Tools and skills: the difference, then Gmail, web design and PPTX; Firecrawl web search; scheduled jobs; Activity 2 · One agent + UX skills build the Kopi Kaki website; Activity 3 · One agent + PPT tool and skill build the Kopi Kaki deck (PP Task 2).	57–96
Topic 3: Hermes Teams — Bots, Group Chats, Kanban & Crons for Marketing and Video	Multiple agents: own tools, skills, model, instructions and context window; hub-and-spoke, task decomposition and subagents; Activity 4 · Hire five named bots, put them in one group chat, get LinkedIn, Facebook and Instagram posts; Activity 5 · One prompt builds a video team and its	97–153

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	Kanban board: research → script → video → review → your approval (PP Task 3); Activity 6 · Crons auto-generate next week's posts and video drafts, still behind your approval; Five principles of AI security.	

Learning Units (approved Course Proposal TPG-2024060313)

LU	Learning unit	Delivered through	K & A	Duration
LU1	Foundations of autonomous AI agents and single-agent LLM applications	Topic 1 history, collaboration, harness and Big Five + Activity 1 always-on assistant + Topic 2 Part A: what an AI agent is, Hermes Agent, Desktop and the company folder, provider and model (Nous free Solar Mini 4 or ChatGPT Plus) and a Telegram bot (QR or BotFather)	K1, K6, A1, A3	5 h
LU2	Agent and multi-agent design for efficiency, with humans in the loop	Topic 2 Parts B–C: voice, tools vs skills, Firecrawl, then Gmail (connector + scheduled brief), web design and PPTX + Activities 2–3 (single-agent website and PPT) + Topic 3 bots and group chats + Activity 4 social media bots	K4, K5, A2, A6	5 h
LU3	Evaluate and improve the effectiveness of an AI-agent workforce in product development	Activity 5 Kanban video team (researcher source checks, review and human approval), Activity 6 content crons, five principles of AI security, evaluation metrics and the optional Paperclip comparison	K2, K3, A4, A5	4 h
Assessment	WA-SAQ (1 h) + PP (1 h)	Day 2, 4:30–6:30 pm	K1–K6, A1–A6	2 h

Scheduled LU allocations: 5 h + 5 h + 4 h, plus 2 h assessment; these include tea. Net instruction/practice is 13 h. Retained 9:30–18:30 timing was confirmed by the trainer.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — three tasks plus written analysis, 1 hour, open book. Task 1: ONE Meta Muse or OpenAI Dots screenshot (Activity 1). Task 2: ONE Hermes Desktop/dashboard screenshot of the professional PPT (Activity 3). Task 3: TWO screenshots — the Hermes Kanban board and the video it produced (Activity 5).
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 4:30 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

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Course Schedule

Day 1 — Human–AI overview → always-on assistants → Hermes Agent setup, tools, skills and schedules (Activities 1–3)

Time	Duration	Topic / Activity	LU	Slides
9:30–9:50	20 min	Welcome, attendance, outcomes and official SWDA AI subscription redemption steps	LU1	1–19
9:50–11:05	75 min	**Topic 1 (Part A) — AI history; what agentic AI is; under the hood (brain-and-body analogy, the agentic loop); humans decide; assistant vs digital worker; five collaboration scenarios (Scenario 2 tried live on WhatsApp); use cases, AI for Science, humanoid elderly care**	LU1	20–40
11:05–11:20	15 min	Tea break (included in scheduled LU time)	LU1	—
11:20–12:00	40 min	**Topic 1 (Part B) — deployment evolution (OpenClaw → Hermes → hosted agents); the Big Five and one briefing slide each (Muse, Dots, Grok Bot, Duet, Cue) with embedded setup videos**	LU1	41–48
12:00–13:15	75 min	Lab 1: Activity 1 — 1A join Meta Muse with the trainer's invite code; 1B set up an OpenAI Dot (Pro/Business Premium); 1C speak prompts with Wispr Flow; Kopi Kaki schedule check and email draft (PP Task 1)	LU1	49–56
13:15–13:30	15 min	**Activity 1 checkpoint and results review — source accuracy, approval controls, limits and actual stop state**	LU1	49–56
13:30–14:30	60 min	Lunch break	—	—
14:30–15:30	60 min	**Topic 2 (Part A) — meet Kopi Kaki Co.; what an AI agent is (its components); what Hermes Agent is; install Hermes Desktop and point it at the kopi-kaki folder; provider and model (Nous free Solar Mini 4 or ChatGPT Plus); a Telegram bot (Create with QR, or manually with BotFather)**	LU1	57–69
15:30–16:00	30 min	**Topic 2 (Part B) — voice; tools vs skills; tools and skills from Capabilities; Firecrawl web search**	LU2	70–75
16:00–16:15	15 min	Tea break (included in scheduled LU time)	LU2	—
16:15–16:45	30 min	**Topic 2 (Part C) — tools and skills in action: Gmail by Google sign-in plus a daily email summary; web-design skills from skills.sh; the PPTX tool and pitch-deck skill**	LU2	76–84
16:45–17:30	45 min	Lab 2: Activity 2 — single agent: Mia installs UX skills from skills.sh and builds the Kopi Kaki website	LU2	85–90
17:30–18:15	45 min	Lab 3: Activity 3 — single agent: PPT tool + skill build a professional Kopi Kaki deck (PP Task 2)	LU2	91–96
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance	LU2	97

Scheduled time excluding lunch: 480 minutes (8 hours), including tea and any assessment.

Day 2 — Hermes bots in a group chat, a Kanban video team and content crons (Activities 4–6), AI security, optional Paperclip bonus, then assessment

Time	Duration	Topic / Activity	LU	Slides
9:30–9:45	15 min	Day 1 recap and AM attendance	LU2	97
9:45–10:15	30 min	**Topic 3 (Part A) — introduction to multiple agents (each with its own tools, skills, model, instructions and context window); hub-and-spoke, task decomposition and subagents; hire bots with New**	LU2	97–109

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		Agent; group chats		
10:15–11:15	60 min	Lab 4: Activity 4 — five named bots (researcher, LinkedIn, Facebook and Instagram writers, image creator) in one group chat write the launch posts; you approve each	LU2	110–116
11:15–11:30	15 min	Tea break (included in scheduled LU time)	LU2	—
11:30–12:15	45 min	Lab 5: Activity 5A — video skills and tools; one prompt creates the video team and its Kanban board; researcher source checks and script	LU3	117–122
12:15–13:15	60 min	Lunch break	—	—
13:15–14:25	70 min	Lab 5: Activity 5B — video production, voice-over and review; human approval of the exact video via its review link (PP Task 3)	LU3	123–127
14:25–14:55	30 min	Lab 6: Activity 6 — scheduled jobs auto-generate the weekly social posts and a video draft; you approve before anything is used	LU3	128–132
14:55–15:10	15 min	Tea break (included in scheduled LU time)	LU3	—
15:10–15:40	30 min	**Topic 3 (Part B) — five principles of AI security; RAG vs a standard LLM; four metrics to evaluate and improve the team**	LU3	133–138
15:40–16:00	20 min	**Bonus (optional) — Paperclip: Kopi Kaki as a one-person company (trainer demo; full lab self-paced) and comparison with the Hermes Kanban team**	LU3	139–153
16:00–16:15	15 min	Course recap, practice questions and Q&A	LU3	154–156
16:15–16:30	15 min	**Briefing for Assessment**	LU3	16
16:30–17:30	60 min	**Written Assessment (WA) — 1 hour, open book; TRAQOM survey and assessment attendance first**	Assessment	157–160
17:30–18:30	60 min	**Practical Performance (PP) — 1 hour, open book; submit answers on the LMS and take PM attendance**	Assessment	157–160

Scheduled time excluding lunch: 480 minutes (8 hours), including tea and any assessment.

Lab Reference

Lab	Title	Paradigm	LO	Slides
Lab 1 (Day 1)	Activity 1 — Your Always-On CEO Assistant: Meta Muse or OpenAI Dots	Personal assistant + human decisions	LO1	49–56
Topic 2 Setup (Day 1)	Hire Your First AI Employee — Hermes Agent Setup	One agent, many tools	LO1/ LO2	81–84
Lab 2 (Day 1)	Activity 2 — One Agent Builds the Kopi Kaki Website with UX Skills	Single agent + installed UX skills	LO2	86–90
Lab 3 (Day 1)	Activity 3 — One Agent Builds the Kopi Kaki Pitch Deck with a PPT Tool and Skill	Single agent + PPT tool and skill	LO2	93–96
Lab 4 (Day 2)	Activity 4 — Your Social Media Team: Five Bots in One Group Chat	Multiple named bots in one group chat	LO2	112–116
Lab 5 (Day 2)	Activity 5 — Your Video Team on a Kanban Board	Agent team on a Kanban board + human gates	LO3	123–127
Lab 6 (Day 2)	Activity 6 — Crons Make Next Week's Posts and Video While You Sleep	Scheduled jobs that start the team's work	LO3	129–132
Bonus Lab (Day 2)	Paperclip: Run Kopi Kaki as a One-Person Company	AI company with an org chart, governed by you	LO3	149–153

Lab materials: one folder per lab under labs/ (README, LAB-<code>-Instructions.pdf, COPY-PROMPTS.txt, data/) and the Lab Prompts page prompts/index.html (also on the course GitHub Page). The bonus Paperclip lab is optional.

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